



POLITECNICO
MILANO 1863

SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE

Requirements Analysis and Specification Document

SOFTWARE ENGINEERING FOR AUTOMATION — RACE MANAGER

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Academic Year: 2025–26

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1 | Introduction

1.1. Scope

Race Manager supports the preparation, supervision, and review of kart racing events. Drivers register individually or as part of a team, submit event waivers, consult their participation status, and review race information. Race directors organize events and manage race operations. Administrators manage accounts, shared configuration, signed documents, and official result imports. Viewers follow live timing.

The system boundary includes the mobile application and its backend service. Physical karts, timing transponders, trackside personnel, third-party timing platforms, and any external payment process remain outside that boundary. Race Manager observes externally supplied timing information; it does not measure lap times directly or physically enforce flags and penalties.

The initial scope includes:

- Accounts, authentication, and differentiated access.
- Circuit and event information, individual and team registrations, and waiting lists.
- Event waiver submission and administrative document retrieval.
- Live timing, kart assignments, stint tracking, race control messages, and penalties.
- In-app notifications, official results, and available performance summaries.

Payment processing, mobile push delivery, Android and web clients, complete offline operation, and advanced telemetry analysis are future extensions. Recording a registration cost or a `pending_payment` status does not imply that the application processes payments. Adding teammates by email does not imply an implemented email delivery or invitation acceptance service.

1.1.1. World phenomena

ID	Phenomenon outside the software boundary
W1	Drivers and teams participate in a physical kart race.
W2	Karts complete laps and cross timing detection points.

W3	Drivers enter the pits and change drivers during endurance races.
W4	Race officials decide when to start, interrupt, resume, or stop a session and whether to impose penalties.
W5	Organizers check participation conditions and any payments outside the application.
W6	A participant reads and agrees to the conditions of an event waiver.

1.1.2. Shared phenomena

ID	Controller	Phenomenon visible across the system boundary
S1	World	A person submits account details and authentication credentials.
S2	World	A participant submits or changes an individual or team registration.
S3	World	An official submits event settings, registration decisions, kart assignments, pit changes, penalties, or race commands.
S4	World	A participant submits personal details and a drawn signature for an event waiver.
S5	World	A timing provider exposes standings and lap timing data.
S6	World	An administrator supplies an official results file or changes shared configuration.
S7	Machine	The application displays event details, participation status, and operation outcomes.
S8	Machine	The application displays timing updates, stint information, messages, and penalties.
S9	Machine	The application makes notifications and signed waiver documents available to authorized users.
S10	Machine	The application presents published results and available performance summaries.

Internal phenomena include validating requests, maintaining records, calculating stint durations, normalizing provider data, and distributing updates to the correct subscribers.

1.2. Purpose

This document provides a common reference for implementation work and acceptance activities. Goals describe the expected outcomes; requirements describe observable software obligations; domain assumptions identify conditions outside the software's control.

Goal	Intended outcome	Relevant phenomena
G1	Participants can prepare for an event and understand their eligibility and registration status.	W1, W5, W6, S1, S2, S4, S7, S9
G2	Team representatives can organize team participation and follow team performance.	W1, W3, S2, S3, S7, S8
G3	Viewers and participants can follow current race progress.	W2, S5, S8
G4	Race directors can coordinate event and race operations using consistent information.	W3, W4, S3, S5, S7, S8
G5	Administrators can maintain accounts, permissions, circuits, and event documents.	S1, S4, S6, S9
G6	Participants can consult published results and review available performance data.	W2, S5, S6, S10
G7	Users receive understandable feedback and appropriately restricted access throughout the event lifecycle.	S1–S10

1.3. Definitions, Acronyms, and Abbreviations

Term	Definition
RASD	Requirements Analysis and Specification Document.
Actor	A person or external system interacting with Race Manager.
Event	An organized kart racing activity with a schedule, participation rules, and registrations.

Session	A named on-track activity within an event, such as qualifying or a race.
Registration	The association between a participant and an event, optionally including team membership.
Team representative / leader	The participant responsible for a team's registration; not a separate global permission role.
Stint	An interval of on-track driving tracked for an assigned kart, excluding pauses and pit time according to the rules below.
Waiver / release form	Event conditions and participant information submitted with a drawn signature.
Official result	A result published through an authorized administrative operation.
Live timing	Continuously refreshed information obtained from an external timing source; not necessarily the final official classification.
API / REST	Application Programming Interface / Representational State Transfer.
JSON	Javascript Object Notation used for internal data exchange.
JWT	JSON Web Token used for authenticated access.
HTTPS / WSS	Encrypted HTTP / encrypted WebSocket communication.
CSV / PDF	Comma-separated values / Portable Document Format.
UML	Unified Modeling Language.
Alloy	A language and analyzer for bounded relational modeling.
TBD	To be decided or confirmed by the project stakeholders.

Identifiers R, P, and Q denote functional, performance, and quality requirements. “Shall” expresses a specification obligation. A requirement is not a claim of existing implementation.

1.4. Document Structure

- **Section 2** describes actors, domain concepts, lifecycle rules, and assumptions.
- **Section 3** defines scenarios, use cases, interface requirements, functional requirements, quality requirements, and traceability.

2 | Overall Description

2.1. Product Perspective

Race Manager combines a native iOS client with a backend that maintains event records and retrieves timing information from supported providers, such as ApexTiming and RaceFacer. Users interact through role-appropriate screens. Several users may follow the same timing source, while others follow a different source.

Production access uses a configured remote service. Local discovery supports development use. Neither network discovery nor access to a timing page grants administrative privileges.

2.1.1. Domain model

Concept	Relevant information and relationships
User	Identity, credentials, profile, and one global role; may have registrations in multiple events.
Circuit	Track information and a supported timing source; may host multiple events.
Event	Schedule, circuit/location, registration limits, team-size limits, waiver text, event status, and operational session status.
Registration	Belongs to one event and identifies a participant by an account or a pending email association; belongs to one event-specific team.
Team	Conceptual grouping of registrations within one event, with a leader and a name.
Signed waiver	Associates a participant with an event and the submitted personal information and signature.
Kart assignment	Associates a kart number within an event with an individual entry or team.
Penalty	Refers to an event and kart, with type, optional time value, note, and timestamp.

Race message	Refers to an event and optionally a particular kart; includes type, text, and timestamp.
Timing snapshot / lap	Provider-derived information associated with a source session and, where available, an event and kart.
Result	Event classification data, optional participant/team association, timing values, and official standings indicator.
Notification	A message belonging to a recipient, optionally linked to an event, with read status.

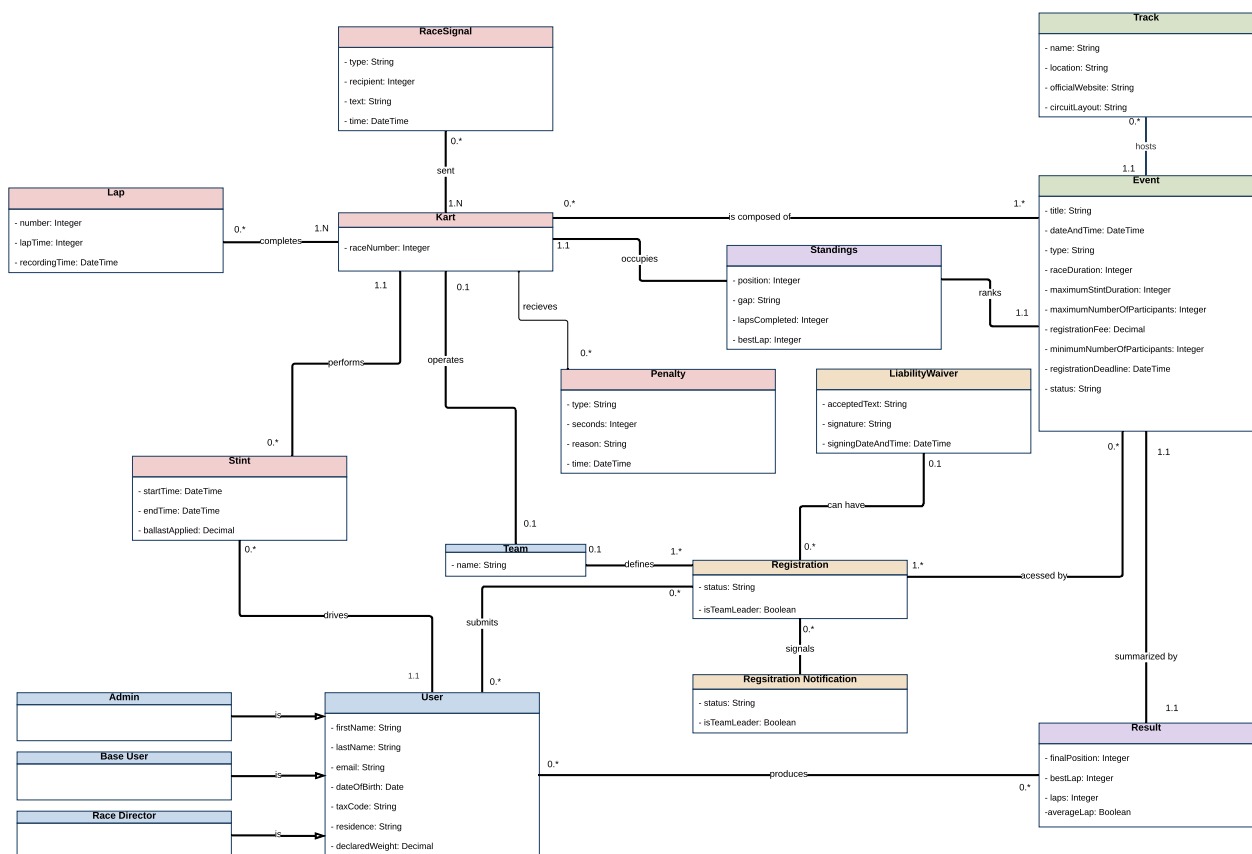


Figure 2.1: Domain class diagram

2.1.2. Event and session lifecycle

Event lifecycle and operational session state are separate. Finishing an event must not be confused with pausing a session or publishing results.

Lifecycle	State	Meaning
Event	scheduled	The event is planned.
Event	started	The event is underway and may contain multiple sessions.
Event	finished	The event is closed; historical information remains available.
Session	not_started	No operational session has begun.
Session	running	Eligible on-track stint timers advance.
Session	paused	Timers are suspended and their elapsed values are preserved.
Session	stopped	The session has ended.

The intended normal event progression is **scheduled** → **started** → **finished**. A scheduled event may also close through the expiration policy. An explicit start action initializes stint counters and starts the session. Yellow and blue flags can be signaled, a green flag clears their effects, and a checkered flag or a red flag stops it. A subsequent session requires an explicit new start.

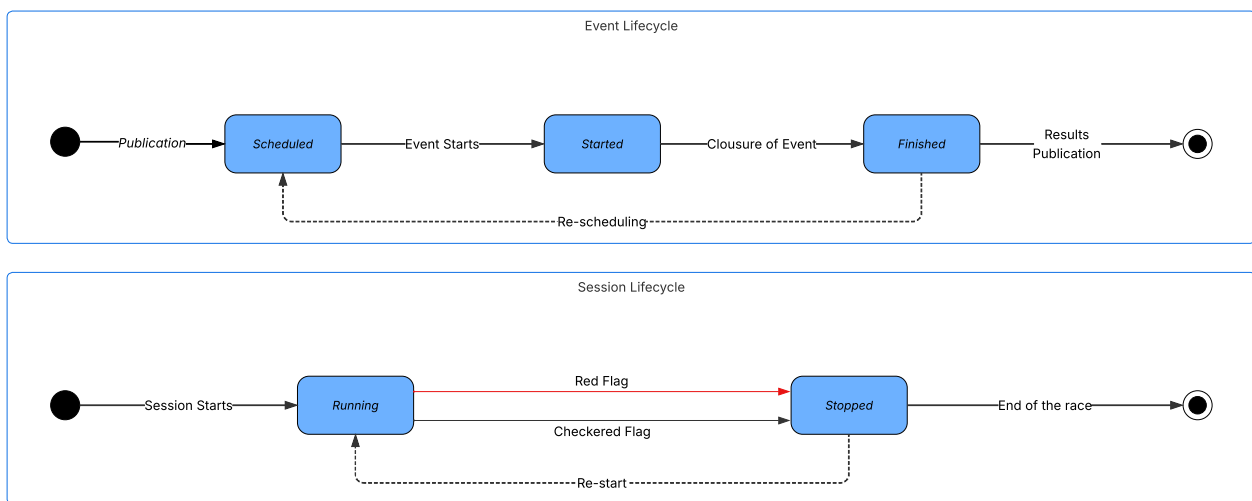


Figure 2.2: Event and Session lifecycles

2.1.3. Registration lifecycle

State	Meaning	Intended transitions
pending_payment	Entry is awaiting organizer confirmation, including any external payment check.	Confirm to confirmed ; move to waitlist ; eligible cancellation.
waitlist	Entry is awaiting admission.	Admit to pending_payment ; eligible cancellation.
confirmed	Organizer has confirmed the entry.	Organizer may return it to pending_payment or remove it.

Late registrations and entries exceeding admission capacity enter the waiting list. An individual requesting participation in a team-format event without a team also enters the waiting list. Ordinary self-cancellation of a confirmed registration is blocked in the current workflow; leaving a team has a separate policy. Pending teammate account association is distinct from registration status. “Invited” and “rejected” are not baseline registration states.

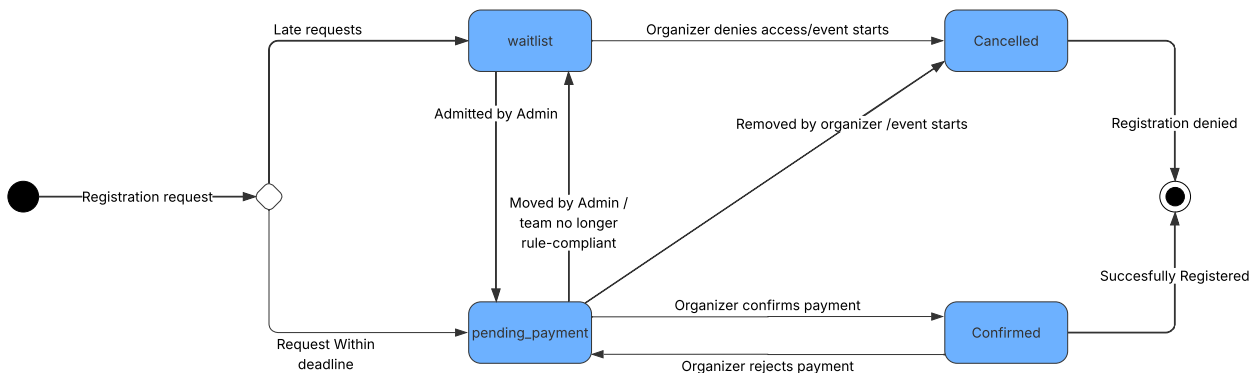


Figure 2.3: Event registration lifecycle

2.2. Product Functions

Actor	Principal functions
Visitor	Access just live timing data from available circuits. Doesn't require signing in and hasn't access to other functions.
Authenticated viewer	Create an account and sign in. Has access to events' informations and can register to them.
Driver	Maintain a profile, register, sign an event waiver, see their kart/team and notifications, and consult results.
Team representative	Create and maintain the team's entry, identify teammates, and follow team race information.
Race director	Manages all live event related actions, such as update pit status, control sessions, deliver messages and penalties.
Administrator	Perform inherited operational functions and manage user roles, circuits, waiver administration, and official result imports.
Timing provider	Expose the externally measured timing information retrieved by Race Manager.

Application roles are ordered `viewer` < `user` < `race_director` < `admin`. A team leader is identified within an event rather than through an additional global role. The baseline specification reserves participation functions for `user` or higher; consistent enforcement for `viewer` accounts requires review against current endpoints. The `admin` role grants the highest privileges and functionality, with additional capabilities not available to other accounts.

2.3. User Characteristics

Drivers and viewers may have little technical knowledge and use the application briefly in a busy trackside environment. They need readable information and explicit outcomes. Team representatives need to acquire timing and stint information quickly. Race directors need efficient controls and protection against accidental state changes. Administrators need searchable records and precise permission boundaries.

Users are expected to understand basic karting terms, while the interface should explain application-specific statuses such as waiting-list admission and pending confirmation.

2.4. Assumptions, Dependencies, and Constraints

2.4.1. Dependencies and constraints

- The initial client is an iOS application.
- The existing implementation uses SwiftUI, FastAPI, SQLite, and browser-based provider adapters.
- Timing depends on reachable provider pages with interpretable data. Provider changes can require adapter maintenance.
- Remote use depends on network connectivity and the configured service endpoint; local discovery is a development facility.
- Waiver export requires PDF generation. Official result import currently requires an administrator-supplied CSV file.
- The application has no baseline integration with payment gateways, transponders, or physical flag equipment.

2.4.2. Domain assumptions

ID	Assumption
D1	Organizers supply the correct schedule, participation rules, waiver text, and circuit information.
D2	A provider's timing values correspond to the selected physical session and are accurate enough for the intended display.
D3	Officials record kart assignments and pit activity correctly when these are not supplied automatically.
D4	Participants provide their own accurate personal details and sign their own event waiver.
D5	Organizers verify any payment externally before confirming a registration where payment is required.
D6	An administrator grants elevated roles only to people authorized by the organizer.
D7	Devices and the backend have usable connectivity during live operation; disconnection remains a handled exception.
D8	The backend clock is sufficiently accurate for registration deadlines and elapsed-time calculations.
D9	Officials remain responsible for physical race decisions and enforcement outside the application.

3 | Specific Requirements

3.1. Scenarios Identification

SC1 — Preparing for an individual race. User creates an account, finds a scheduled sprint race, and reads the cost and deadline. Her registration becomes pending confirmation. She reads and can sign the event waiver. After checking the external payment, the director confirms her registration, and User receives an in-app notification.

SC2 — Organizing an endurance team. User creates a team and supplies his teammates' identifiers. The system validates membership and team size. Existing users see the relevant notifications; an unlinked email/username remains distinguishable from a registered account. User follows the team's registration.

SC3 — Joining a full event. User requests entry after available capacity has been reached. The system places him on the waiting list. A director later admits him to pending confirmation and confirms him only after the necessary organizer checks.

SC4 — Following a live timing with a feed interruption. User selects a timing source and receives standings updates. When the provider or connection becomes unavailable, the application indicates the interruption. Following recovery, it displays fresh data for the selected source.

SC5 — Controlling an interrupted endurance session. Race Director starts a session, records pit entries, and sends a red flag. Active stint timers pause. A stopped session cannot resume through a green flag alone but has to be re-initiated.

SC6 — Applying a stint penalty. An assigned kart exceeds the configured stint limit. The backend records one automatic penalty for that stint, even if no mobile client is connected. A Race Director can review the penalty; deleting it does not cause the same stint to generate it repeatedly.

SC7 — Publishing results. An administrator imports a classification CSV. The system reports accepted rows and row-level problems. If every row is invalid, it preserves the previous classification. Participants can subsequently inspect the official results and available lap statistics.

SC8 — Updating administrative access. An administrator promotes a user to race director. Subsequent server-side authorization uses the current role. Only administrators can change other users' privileges.

3.2. UML Diagrams Analysis

3.2.1. Use case diagram

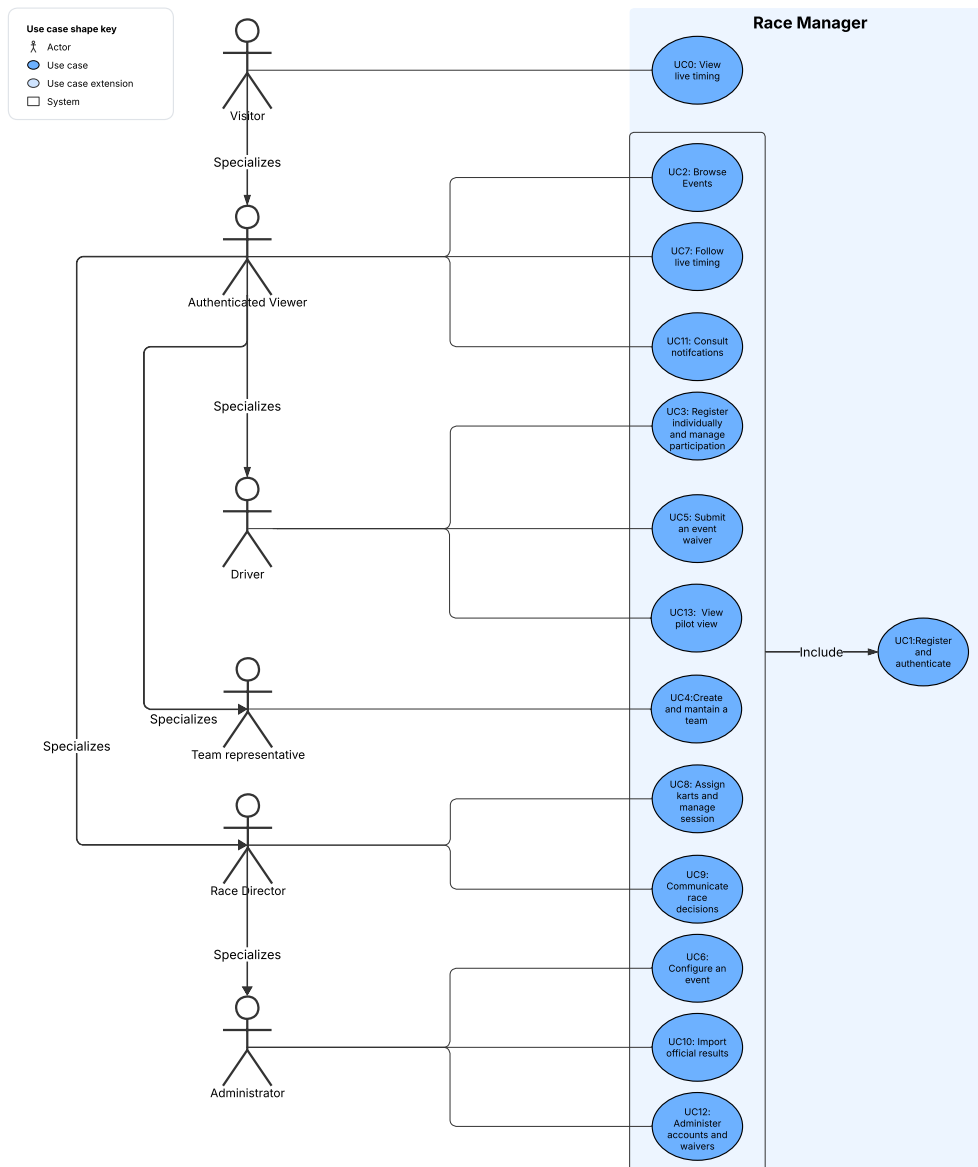


Figure 3.1: Use case diagram

3.2.2. Use case identification

The successful outcomes below assume that server-side validation and persistence succeed. On failure, the system must not report success; any already committed state must be recoverable through a fresh read.

UC0 — View live timing

- **Actors:** Visitor.
- **Preconditions:** The service is reachable; at least one race session is active or has recently concluded.
- **Trigger:** The person opens the live-timing page without any prior authentication.
- **Main flow:** 1. The system presents the public live-timing view. 2. Lap times, sector times, and gap data are streamed in real time via WebSocket. 3. The view updates automatically as new timing events arrive from the External Timing Provider. 4. The visitor may browse session standings and individual driver lap without interacting with any authenticated endpoint.
- **Postconditions:** The visitor has observed current or recent session data; no account or session token is created.
- **Alternatives/exceptions:** If no session is active, the system displays the most recent archived session with a clear notice. If the WebSocket connection drops, the client attempts automatic reconnection; after repeated failures it falls back to showing the last known snapshot with a connectivity warning.

UC1 — Register and authenticate

- **Actors:** Visitor; registered user.
- **Preconditions:** The service is reachable; login requires an existing account.
- **Trigger:** The person chooses account creation or login.
- **Main flow:** 1. Submit account details or credentials. 2. The system validates them. 3. For registration, it creates a non-privileged account. 4. On successful login, it establishes an authenticated session and shows the appropriate home screen.
- **Postconditions:** A valid account or authenticated session exists.
- **Alternatives/exceptions:** Duplicate identity, invalid credentials, or invalid input produces a clear failure. An expired session requires refresh or renewed authentication. Logout ends the local session and revokes the relevant refresh token.

UC2 — Browse events and inspect an event

- **Actors:** Authenticated viewer, driver, director, administrator.
- **Preconditions:** Event browsing is accessible to the actor.
- **Trigger:** Open the event list or an event-linked notification.
- **Main flow:** 1. Obtain the event list. 2. Select an event. 3. Read its schedule, format, rules, cost, status, and available participation information.
- **Postconditions:** The actor can identify the event and relevant next actions.
- **Alternatives/exceptions:** An empty list, deleted event, or unavailable service has a distinct explanatory state.

UC3 — Register individually and manage participation

- **Actors:** Driver.
- **Preconditions:** Authenticated participation access; an existing event; no duplicate registration.
- **Trigger:** Request event registration or cancellation.
- **Main flow:** 1. Select the event. 2. Submit the required participation information. 3. The system checks format, deadline, duplication, and capacity. 4. It creates the registration in pending confirmation or on the waiting list and displays the outcome.
- **Postconditions:** One registration represents the participant's request for that event.
- **Alternatives/exceptions:** A full event, late request, or unassigned individual in a team event enters the waiting list. Duplicate requests are rejected. Eligible cancellation removes the request; cancellation of a confirmed entry is blocked.

UC4 — Create and maintain a team

- **Actors:** Team representative; team member; director for administrative changes.
- **Preconditions:** A team-format event; appropriate operational authority.
- **Trigger:** Create/edit a team or leave a team.
- **Main flow:** 1. Supply a team name and teammate identifiers. 2. Validate size and duplicate membership. 3. Create or update the event-specific group and leader association. 4. Notify linked accounts and display the resulting membership.
- **Postconditions:** Team membership is stored consistently within the event.
- **Alternatives/exceptions:** Unknown accounts remain pending email associations where supported. An unauthorized edit or duplicate member is rejected. A non-leader may use the separate leave-team action.

UC5 — Submit an event waiver

- **Actors:** Driver.
- **Preconditions:** Authenticated identity, existing event, and access to the applicable waiver text.
- **Trigger:** Open the event waiver form.
- **Main flow:** 1. Read the waiver. 2. Enter required personal details. 3. Draw a signature and preview the document. 4. Submit. 5. The system associates the signed data with the participant and event and displays signed status.
- **Postconditions:** The signed record can be retrieved by authorized actors.
- **Alternatives/exceptions:** Incomplete details or storage failure prevents success confirmation. Administrative removal or invalidation is reflected in the participant's status.

UC6 — Configure an event and manage registrations

- **Actors:** Administrator.
- **Preconditions:** Operational authority.
- **Trigger:** Create/edit an event or review its registrations.
- **Main flow:** 1. Configure event details and participation rules. 2. Review individual and team entries. 3. Admit eligible waiting-list entries to pending confirmation. 4. Confirm entries after external organizer checks. 5. Notify affected participants.
- **Postconditions:** Event settings and registration decisions are available consistently.
- **Alternatives/exceptions:** Invalid limits or state transitions are rejected. Reversing confirmation returns an entry to pending confirmation. Team actions must produce coherent updates rather than partially changed membership statuses.

UC7 — Follow live timing

- **Actors:** All application users; external timing provider.
- **Preconditions:** A supported timing source is available for selection.
- **Trigger:** Select a source or open the live timing view.
- **Main flow:** 1. Authenticate the connection. 2. Select an allowed source. 3. Retrieve and normalize timing data. 4. Distribute snapshots to subscribers of that source. 5. Display available positions, kart identifiers, lap values, and gaps.
- **Postconditions:** The client follows only the selected source and can identify its connection/data status.
- **Alternatives/exceptions:** An unsupported source is rejected. Provider failure, expired credentials, disconnection, or capacity exhaustion produces an explicit status. Recovery obtains a fresh snapshot.

UC8 — Assign karts and manage session/pit state

- **Actors:** Race director or administrator.
- **Preconditions:** An existing event and valid entries; operational authority.
- **Trigger:** Assign a kart, start a session, or record a pit/state change.
- **Main flow:** 1. Associate the kart with an entry. 2. Start the session explicitly. 3. Record pit entry/exit as required. 4. Start or stop the session. 5. Distribute the authoritative state to affected clients.
- **Postconditions:** Kart assignments and stint values reflect the recorded operational state.
- **Alternatives/exceptions:** A conflicting kart assignment is rejected. Session end preserves elapsed time; pit exit starts a new stint. Automatic event closure follows the separate event policy.

UC9 — Communicate race decisions and manage penalties

- **Actors:** Race director or administrator; backend stint monitor as an internal initiator.
- **Preconditions:** Existing event and, for a targeted operation, a valid event kart.
- **Trigger:** Submit a message/penalty or detect a stint violation.
- **Main flow:** 1. Select the event, target, and action. 2. Validate the action and applicable rule. 3. Store the message or penalty. 4. Update race state when the action is a global control command. 5. Display messages and accumulated penalty information to relevant users.
- **Postconditions:** The decision is available in the correct event context.
- **Alternatives/exceptions:** Negative penalty durations and kart-targeted global state commands are rejected. Automatic stint assessment records at most one penalty per stint. Authorized penalty removal updates totals without automatically rearming that stint.

UC10 — Import official results

- **Actors:** Administrator.
- **Preconditions:** Existing event and a results CSV available to the administrator.
- **Trigger:** Choose official results import.
- **Main flow:** 1. Select the event and file. 2. Parse and validate rows. 3. Associate results with entries where possible. 4. If at least one row is valid, replace the event's existing official results in one committed operation. 5. Report imported count and row issues.
- **Postconditions:** The accepted official classification is available for consultation.
- **Alternatives/exceptions:** A file with no valid results preserves the previous classification. A partial import explicitly reports skipped/invalid information. Unmatched

results remain identifiable without falsely associating them with a user. The replacement currently affects all official result types for the event.

UC11 — Consult notifications, results, and performance

- **Actors:** Driver, team representative; other authenticated users for permitted result views.
- **Preconditions:** An authenticated session; records available for the selected scope.
- **Trigger:** Open notifications, event results, or personal analysis.
- **Main flow:** 1. Retrieve the actor's notifications or selected results. 2. Open linked event information. 3. Inspect official classification and available lap summaries. 4. Mark personal notifications as read or delete them.
- **Postconditions:** The actor can distinguish official results from live or other performance information.
- **Alternatives/exceptions:** Missing data is shown as unavailable, not as zero performance. Unmatched results are not silently attributed to the actor. Access to another person's private notifications is denied.

UC12 — Administer accounts and waivers

- **Actors:** Administrator.
- **Preconditions:** An authenticated administrator session.
- **Trigger:** Open administrative tools.
- **Main flow:** 1. Find the relevant account or document. 2. Review current data. 3. Apply a role/configuration change or retrieve a signed PDF. 4. Confirm the persisted outcome.
- **Postconditions:** Administrative information is updated or an authorized document is retrieved.
- **Alternatives/exceptions:** Invalid role values, unauthorized access, and missing records are rejected. A removed/inactivated waiver must not continue to appear signed.

UC13 — View pilot view

- **Actors:** Driver.
- **Preconditions:** The driver is authenticated; at least one race session is active or has recently concluded; the driver has been assigned to that session by the system.
- **Trigger:** The driver navigates to their personal pilot-view screen during or immediately after a session.
- **Main flow:** 1. The system verifies the driver's identity and session assignment. 2. It retrieves the driver's real-time telemetry feed for the active session: lap time, gaps and current position. 3. Data is streamed live via WebSocket and the view refreshes

automatically on each new timing event. 4. Any session-relevant message issued by the Race Director such as penalties, warnings, or flag signals, is delivered as a prominent, unmistakable alert that overlays the view.

- **Postconditions:** The driver has observed their own session data in real time; no state change is persisted beyond the session record already captured by the system.
- **Alternatives/exceptions:** If the WebSocket connection drops, automatic reconnection is attempted; after repeated failures the last known snapshot is shown.

3.3. External Interface Requirements

3.3.1. User interfaces

The interface shall provide role-appropriate navigation, clear empty/loading/error states, and visible confirmation of completed changes. Critical race information shall remain readable without relying only on color. Administrative actions shall identify the affected event, team, kart, or user.

View	Required information and interactions
Authentication/profile	Registration, login/logout, profile information, and actionable validation errors.
Event list/detail	Schedule, location, format, limits, cost, status, registration state, and available actions.
Team management	Leader, members, unresolved account associations, size limits, and permitted edits.
Waiver	Readable event text, required personal fields, signature area, preview, submission outcome.
Live timing	Selected source/session, standings, kart identifiers, available lap/gap data, freshness and connection status.
My kart/team	Assignment, available stint/pit state, penalties, and relevant race messages.
Race control	Assignments, session controls, pit changes, messages, penalties, and confirmation of critical commands.
Results/analysis	Official classification, available lap summaries, and explicit missing-data states.
Administration	User/role search, circuits, signed waivers, and result import feedback.
Notifications	Personal messages, read status, and links to the relevant event.

3.3.2. Hardware interfaces

The initial application uses a touch-capable iOS device for navigation and signature input. No dedicated kart or timing hardware interface is required. Physical lap measurement belongs to the timing provider. Optional location-related experiments in the repository are not a baseline requirement of this RASD.

3.3.3. Software interfaces

The mobile client communicates with the backend for accounts, events, registrations, waivers, notifications, live operations, and results. Structured requests and responses use JSON, result import uses a CSV upload, and document retrieval returns PDF content. Provider adapters transform supported timing pages into a common representation; unavailable provider fields must remain distinguishable from real zero values.

The CSV interface shall document accepted headers, units, encodings, required classification fields, and participant matching behavior. Import feedback shall identify problematic rows. Detailed endpoint contracts belong in API documentation and must remain consistent with the requirements here.

3.3.4. Communication interfaces

Production client/server communication shall use HTTPS and WSS. Protected REST actions and live connections require authenticated identity. Timing subscriptions shall remain separated by source; event operation updates shall remain associated with their event. The client shall distinguish its selected source from shared circuit configuration.

Local development may use Bonjour/mDNS discovery and an explicitly selected local endpoint. Development transport settings shall not silently replace production security settings. Connection loss shall be represented in the UI and handled through reconnect/re-authentication as appropriate.

3.4. Functional Requirements

These requirements define the intended baseline.

3.4.1. Identity and access

ID	Requirement
R1	The system shall create non-privileged accounts from valid registration details and reject duplicate usernames or email addresses.
R2	The system shall authenticate credentials, support session renewal, and support logout with refresh-token revocation.

R3	The system shall authorize protected actions on the server using the current account role and applicable ownership/membership rules; invalid or expired identities shall not retain protected live access.
R4	The system shall allow authenticated users to view and update supported personal profile fields without changing their own global role.

3.4.2. Events and registrations

ID	Requirement
R5	The system shall list events and expose their schedule, location, format, participation limits, registration cost where specified, and lifecycle status.
R6	The system shall show a participant's registration and waiver status separately for each event.
R7	The system shall allow race directors and administrators to create, update, and remove events, validating supplied dates and limits and maintaining consistent dependent records.
R8	The system shall allow eligible participants to register individually and shall prevent multiple registrations for the same linked account in one event.
R9	The system shall route late requests, requests beyond configured capacity, and individual requests awaiting a team in a team-format event to waitlist ; other admissible requests shall enter pending_payment .
R10	The system shall display registration outcomes and allow eligible self-cancellation while blocking ordinary self-cancellation of confirmed entries.
R11	The system shall allow an eligible participant to create an event-specific team with a designated leader, a name, and member identifiers, validating configured team-size limits and duplicate identities.
R12	The system shall restrict participant team edits to the relevant leader and provide a separate non-leader departure operation; administrative changes require operational authority.
R13	The system shall let race directors and administrators admit waiting-list entries, confirm pending entries, reverse confirmation, and remove entries, applying team-wide operations consistently.
R14	The system shall report invalid registration transitions without partially applying them and shall retain an understandable distinction between membership, confirmation, and signed-waiver status.

3.4.3. Waivers

ID	Requirement
R15	The system shall present the applicable event waiver text and allow the participant to preview the document before submitting required personal details and a drawn signature.
R16	The system shall associate a successfully submitted waiver with its participant and event and display signed status only after successful storage.
R17	The system shall provide authorized administrative consultation and PDF retrieval of signed waivers and shall reflect removal/invalidation in the participant's status.

3.4.4. Live timing

ID	Requirement
R18	The system shall allow authenticated viewers and higher roles to subscribe to an allowed supported timing source without altering another client's source selection.
R19	The system shall retrieve and normalize available provider standings, including kart identifiers, positions, lap values, and gaps where supplied.
R20	The system shall distribute each timing snapshot only to clients subscribed to its source and distribute event updates within the correct event context.
R21	The system shall identify unavailable/stale timing or disconnected service state and obtain a fresh snapshot on recovery instead of presenting retained information as current.

3.4.5. Race operations

ID	Requirement
R22	The system shall allow authorized officials to associate event kart numbers with entries and shall reject conflicting simultaneous assignments of the same kart within an event.
R23	The system shall maintain event lifecycle separately from session operational state and require an explicit start action for a new session after stop.

R24	The system shall advance stint time only while the session is running and the assigned kart is on track; pause and pit entry shall preserve elapsed time, while pit exit begins a new stint.
R25	The system shall make the participant's relevant kart assignment, stint/pit state, penalty total, and applicable messages available for consultation.
R26	The system shall let authorized officials record broadcast or kart-targeted messages; commands changing global session state shall not accept a single-kart target.
R27	The system shall let authorized officials record and remove penalties with a valid type, optional non-negative time value, and explanatory note, and shall update displayed totals accordingly. Configured warning thresholds shall trigger their specified automatic penalty.
R28	The system shall evaluate configured stint limits on the backend, issue at most one automatic stint penalty per stint, and avoid reissuing that penalty merely because it was deleted. A new stint resets eligibility for assessment.
R29	Under the provisional expiration policy, the system shall close scheduled or started events 48 hours after their scheduled start and notify connected clients of the lifecycle update; this shall not constitute result publication.

3.4.6. Results, notifications, and administration

ID	Requirement
R30	The system shall restrict official CSV import to administrators, validate rows before replacing official event results, report partial import issues, and preserve the previous classification when no valid rows are available.
R31	The system shall provide permitted event results, personal result history, and available lap summaries, distinguishing official data and unavailable participant associations.
R32	The system shall provide recipient-specific in-app notifications for relevant registration/team/waiver changes and reminders, with read/delete operations restricted to the recipient and event links where applicable.
R33	The system shall allow administrators to search accounts and assign supported roles; ordinary users shall not perform these changes.
R34	The system shall allow administrators to maintain circuit information, supported timing-source configuration, and shared waiver settings.

3.4.7. Traceability

Goal	Requirements	Use cases	Main assumptions
G1	R1, R2, R4–R10, R13–R17, R32	UC1, UC2, UC3, UC5, UC6, UC11	D1, D4, D5
G2	R11–R14, R22, R24, R25, R32	UC4, UC8, UC11	D1, D3
G3	R18–R21	UC7	D2, D7
G4	R7, R13, R14, R22–R29	UC6, UC8, UC9	D1, D3, D8, D9
G5	R3, R17, R33, R34	UC12	D4, D6
G6	R30, R31	UC10, UC11	D2, D3
G7	R2, R3, R6, R10, R14, R16, R20, R21, R32	UC1–UC12	D6, D7

3.4.8. Acceptance examples

Check	Given / When / Expected outcome	Coverage
A1	Given an existing registration, submitting the same participant/event again does not create a second entry.	R8
A2	Given a passed deadline, an otherwise valid new request enters the waiting list and is visibly labeled.	R9
A3	Given a participant account, attempting a role change or another team's edit is denied server-side.	R3, R12, R33
A4	Given two clients subscribed to different sources, an update from one source does not appear on the other client.	R18, R20
A5	Given a running stint, stop freezes elapsed time and pit exit starts a new stint.	R23, R24
A6	Given an exceeded stint limit and no mobile clients, backend assessment creates one penalty; repeated assessment and deletion do not recreate it in that stint.	R28
A7	Given published results, importing a file containing no valid rows preserves those results and reports failure.	R30

A8	Given a successfully submitted waiver, its status and authorized PDF retrieval identify the correct participant and event.	R16, R17
A9	Given a stopped session, a green flag does not restart its timers; an explicit new start does.	R23
A10	Given an interrupted provider feed, the interface displays stale/unavailable state instead of claiming current timing.	R21

3.5. Performance Requirements

The following values are proposed targets for stakeholder review, not measured service guarantees.

ID	Proposed measurable target
P1	At least 95% of ordinary event/profile/registration API requests complete within 2 seconds under the reference load.
P2	At least 95% of normalized timing snapshots reach subscribed clients within 1 second of backend acceptance.
P3	The UI indicates lost connectivity or absent expected timing updates within 10 seconds.
P4	A continuous stint-limit violation is assessed within 2 seconds of exceeding the limit during normal backend operation.
P5	A CSV returns an import outcome within 5 seconds after upload completion.

Resource limits shall fail with a clear response rather than silently attaching a user to the wrong source or losing committed operations.

3.6. Design Constraints

3.6.1. Regulatory and organizational policies

The system handles personal information and drawn signatures. Requirements for permitted processing, retention, deletion, and waiver wording must be supplied and approved by the responsible organization before operational release. This document does not establish legal validity of a drawn signature or claim compliance certification.

Access to signed documents and personal details shall follow the role and ownership requirements. Demonstration data and screenshots should use fictional identities.

3.6.2. Hardware and software limitations

The initial release is constrained to the existing iOS/backend deployment. Browser-based data acquisition consumes backend resources and depends on provider availability and page structure. The initial persistence technology requires verification under the agreed concurrent workload. Mobile clients cannot be assumed to remain connected or active in the background.

Consequently, automatic stint assessment belongs to backend behavior, and the timing interface must tolerate missing data. Offline race control and hardware-level lap timing are outside the baseline.

3.7. Software System Attributes

ID	Attribute	Requirement and verification approach
Q1	Availability	A single provider outage shall not prevent unrelated source subscriptions or ordinary event/account operations. Verify by disabling one provider while using another.
Q2	Reliability	Committed registrations, waivers, results, and race decisions shall survive a normal backend restart. Interrupted operations shall not be reported as successful. Verify restart and transaction-failure scenarios.
Q3	Security	Production transport shall be encrypted; passwords shall not be stored as plaintext; authenticated identity and permissions shall be enforced on the server. Verify direct unauthorized requests in addition to UI restrictions.
Q4	Privacy	Private notifications and waiver records shall be accessible only to their permitted recipient/owner or authorized administrative role. Verify cross-account and direct-document access attempts.
Q5	Usability	Important statuses shall have text labels, and invalid actions shall provide an understandable recovery step. Review the participant and director workflows, including errors and empty states.
Q6	Maintainability	A new timing adapter shall map to the common timing contract without changing registration or waiver behavior. Verify adapter isolation and regression coverage for those independent workflows.
Q7	Portability	Domain operations shall be exposed through documented service interfaces so future clients can reuse them.
Q8	Consistency	Dates and elapsed times shall have explicit units, and updates shall preserve event, participant, and source identity. Verify deadline boundaries and simultaneous independent events.